

**PUBLISHED**

| Question | Answer                                                                       | Marks       |
|----------|------------------------------------------------------------------------------|-------------|
| 3(d)     | $(\max) E_{(P)} / 130 = \frac{1}{2} F_{(0)} x_{(0)}$                         | <b>C1</b>   |
|          | $(F_0 =) 2 \times 130 / 0.37 = 700 \text{ (N)}$                              | <b>A1</b>   |
|          | <b>or</b> $(\max) E_{(P)} / 130 = 1 / 2 k x_{(0)}^2$<br>$(F_0 =) k x_{(0)}$  | <b>(C1)</b> |
|          | $(F_0 =) 2 \times 130 \times 0.37 / (0.37)^2 = 700 \text{ (N)}$              | <b>(A1)</b> |
| 3(e)(i)  | $(\text{weight}) = 5.5 \times 9.81$                                          | <b>C1</b>   |
|          | resultant force = $700 - (5.5 \times 9.81)$<br>= 650 N                       | <b>A1</b>   |
| 3(e)(ii) | $F = ma$                                                                     | <b>C1</b>   |
|          | $a = 650 / 5.5$ <b>or</b> $(700 / 5.5) - 9.81$<br>$a = 120 \text{ m s}^{-2}$ | <b>A1</b>   |